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A Novel Approach to E-Commerce Customer Churn Prediction: Leveraging Random Forest and Extreme Gradient Boosting

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Abstract*:*
This study investigates the intense competition within the e-commerce industry and the high, often fluctuating, customer churn rates that e-commerce platforms experience. A new model has been introduced, utilizing both the random forest algorithm and the extreme gradient boosting algorithm. This model begins by analyzing the factors leading to customer churn through a regression decision tree module and then enhances predictive accuracy by integrating an extreme gradient boosting algorithm. The findings suggest that customer churn on e-commerce platforms is predominantly influenced by product quality and after-sales service. Enhanced after-sales service and superior product quality were linked to lower customer churn rates. The developed model showed improved performance, with a root mean square error approximately 1.6% lower than the best existing model, an accuracy rate of 95.2%, and a recall rate 6.8% higher than the lowest performing model. Furthermore, the model's training time remained stable at around 290 seconds, reflecting increased efficiency and faster prediction capabilities compared to other methods. Consequently, this new model is more effective in predicting customer churn trends on e-commerce platforms.

Keywords: Customer churn; Extreme gradient boosting algorithm; Random Forest algorithm; E-commerce; Accuracy.

1. Introduction

In the digital age, e-commerce faces significant challenges, particularly in areas like finance, credit, and logistics. Logistics, being a crucial component of e-commerce competitiveness, plays a vital role in driving the industry's growth by ensuring stable and efficient operations. As e-commerce has gained prominence in recent years, the importance of logistics services has increased, making them a key factor in the sector's ongoing development. For consumers, the rapid expansion of e-commerce offers more choices, which in turn accelerates the natural selection process within the industry. Consequently, evaluating and predicting customer churn and associated risks on e-commerce platforms has emerged as a critical area of research.

The Extreme Gradient Boosting (XGBoost) algorithm, a widely used machine learning technique, is frequently employed in training frameworks and constructing data classification models. Due to its reliance on Regression Tree (RT) models, XGBoost performs well in data prediction tasks. Similarly, the Random Forest (RF) algorithm is popular for data classification and RT model construction because of its ability to generate highly accurate classifiers. However, traditional methods often fall short in achieving precise customer churn prediction and evaluation.

To address this, the study focuses on evaluating and analyzing Customer Churn Risk (CCR) in e-commerce platforms and introduces a novel model that combines the XGBoost algorithm with the RF model. This new model is capable of performing early warning analyses on the likelihood of customer churn. By leveraging machine learning techniques, this approach enables a deeper understanding of the factors contributing to customer churn and produces a high-precision churn risk assessment model, thereby offering valuable insights for the e-commerce industry.

The evaluation of Customer Churn Risk (CCR) is vital for the operation of e-commerce platforms and their sales strategies. Consequently, much research focuses on understanding and analyzing customer churn on these platforms. For example, Manohar Mishra et al. introduced an island detection technique using a dual tree complex discrete wavelet transform combined with the XGBoost algorithm to capture voltage and current signals from distributed energy resources. This method effectively processed nonlinear sampling data by gathering feature vectors, significantly improving island detection accuracy and sensitivity. Similarly, Li et al. developed an enhanced XGBoost algorithm based on graph isomorphic networks for detecting the adsorption properties of metals, directly extracting feature data from atoms and ensuring more accurate predictions of metal adsorption characteristics. Binson et al. proposed a method employing the extreme XGBoost algorithm to enhance the detection accuracy of harmful gases in instruments, which in turn improved traditional diagnostic tools for respiratory diseases. Bhattacharyya et al. conducted an analysis of customer churn in the telecommunications industry, identifying that factors like data transmission, relationship establishment, and server coverage are key contributors to churn. In another study, Han et al. found that e-commerce has the potential to improve poverty alleviation in China. They developed a comprehensive opinion adoption model to analyze the impact of e-commerce on rural purchasing power, showing how the model could enhance farmers' purchasing intentions and improve poverty alleviation efforts through e-commerce.

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Kim et al. introduced a model for analyzing customer churn in the telecommunications sector, focusing on factors like program and financial costs, which significantly enhanced customer retention strategies. Veerabhadraswamy et al. presented a new RF-based sensor data classification method, improving sensor data classification performance by over 13.5%. Zhang et al. also proposed an RF-based approach for detecting rock performance, analyzing characteristics such as permeability and porosity of carbonates, demonstrating a strong generalization effect in predictions. Machine learning, particularly in customer churn analysis, has proven effective not only in data testing but also in prediction within e-commerce. For instance, Gurung et al. constructed a customer churn prediction model using random forests and decision trees to help U.S. companies develop effective retention strategies, achieving high accuracy (96.25%) and uncovering key churn factors. Manzoor et al. synthesized findings from 212 articles on machine learning in customer churn prediction, emphasizing the importance of considering multi-dimensional information and profitability assessment, while also suggesting ensemble learning and deep learning for future research.

Imani et al. focused on improving customer churn prediction accuracy in the telecom industry by combining multiple machine learning models with data sampling strategies and hyper-parameter optimization. Their study found that CatBoost, after optimization, achieved a 93% F1 score, while XGBoost with SMOTE and Tomek Links yielded a 91% ROC AUC, addressing data imbalance issues and significantly enhancing model performance.

In summary, while current research on CCR prediction is limited, it typically focuses on identifying the causes of customer churn without thoroughly analyzing churn data. However, the models used in these studies demonstrate considerable practical capabilities, particularly in data prediction. Employing RF and XGBoost algorithms is therefore feasible for CCR prediction. By using the RF algorithm for feature selection, the most relevant feature set can be identified to evaluate and predict customer churn. Additionally, analyzing data feature importance through model training can help identify key factors leading to customer churn, thereby improving customer service quality on e-commerce platforms and increasing their profitability and market share.

2. Churn Prediction Model – Ecommerce

This section focuses on constructing a Customer Churn Risk (CCR) model for the current e-commerce platform using Random Forest (RF) and Extreme Gradient Boosting (XGBoost) algorithms. Initially, various common methods for evaluating CCR are analyzed. Following this, a new model is developed by enhancing traditional approaches with these advanced algorithms. In the analysis of e-commerce platforms, customer churn is a common problem in the current operation of e-commerce. Due to the complexity of customer churn, traditional data analysis methods and probability models can no longer meet the current CCR prediction for e-commerce platforms [16] . For this reason, the data mining method shown in Figure 1 is commonly used in the mining of e-commerce data.

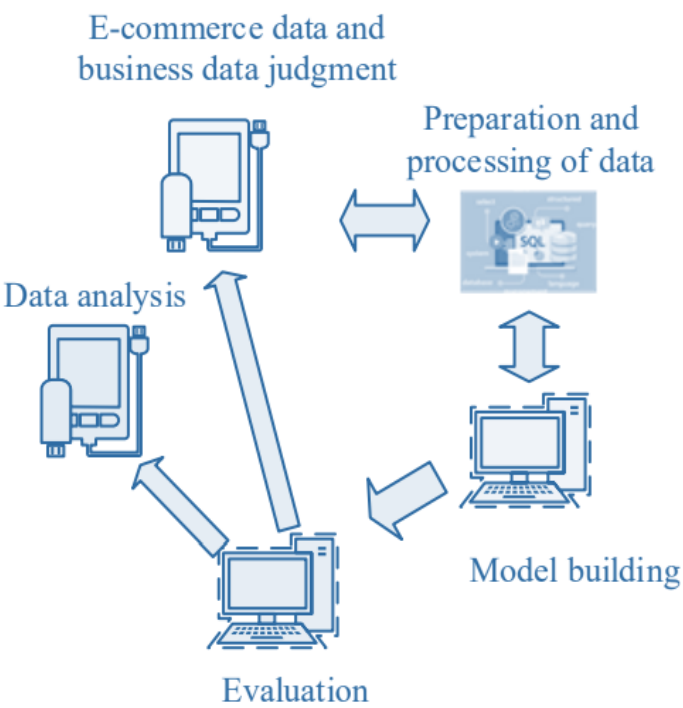

Figure 1: Generic data Mining Model

In Figure 1, the process of analyzing and mining e-commerce data begins by determining whether the current e-commerce and commercial data have potential value that can be assessed and utilized. Based on this value, the data is prepared and processed accordingly. The next step involves constructing a robust data model, which is then evaluated and analyzed to complete the data analysis process. During data analysis, machine learning models are typically used to examine feature datasets. As such, the model-building process relies heavily on data, with feature engineering serving as the framework

for the entire model. Machine learning algorithms are subsequently employed to construct the model. Figure 2 illustrates this process of building a data model. In Figure 2, the model requires a thorough feature engineering analysis of the data to effectively assess the value within e-commerce. This engineering analysis comprises three key components: feature processing, feature selection, and feature analysis. The construction of the model primarily involves handling classification tasks, particularly with imbalanced data, and selecting the appropriate model for the analysis. The results generated by the final model need to be applied precisely to e-commerce data platforms.
Given that e-commerce platforms host multiple brands, each with its own approach to evaluating customer churn, it is essential to tailor the churn assessment to each specific platform.

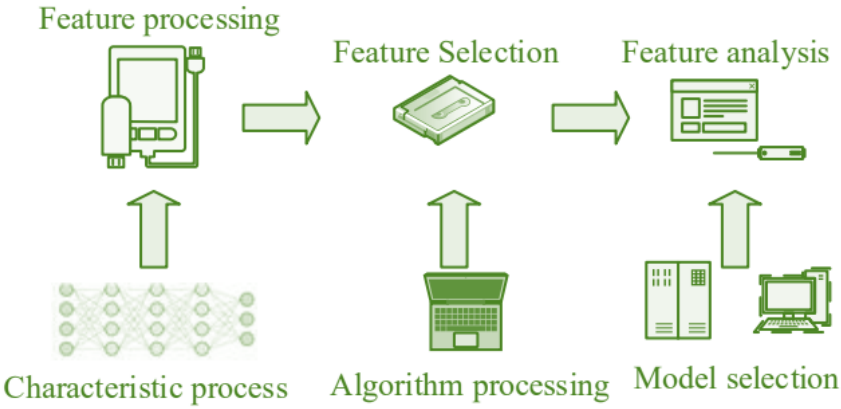

Figure 2: Process involved in data Mining

This is achieved through user survival analysis, which predicts and evaluates the likelihood of a particular event, such as customer churn. Because customer churn on e-commerce platforms can be unpredictable, the evaluation criteria are often based on users' recent purchase volume and the intervals between their purchases. Equation (1) presents the formula used to determine customer churn on e-commerce platforms.

$$Customer\ purchasing\ Status \begin{cases} No\ churn \\ churn \end{cases} \ldots\ldots\ldots Eqn\ 1$$

In equation (1), nochurn and churn represent two types of customer churn states: those that have not yet occurred and those that have occurred. The criterion for judging nochurn is that when the customer's recent purchase level is ≤ the maximum purchase interval, it indicates that there is no loss status. churn indicates that customer churn occurs when the customer's recent purchase level is > the maximum purchase interval. The purchasing status of customers can be determined by describing the probability of an individual's duration. Assuming the current customer's purchase duration is T , the survival function of the customer at this time is expressed as equation (2)

$$R(t) = P(U > t) = \int_t^{\infty} g(u)du = 1 - G(t) \ldots\ldots Eqn\ 2$$

- Let R(t) represent the survival function or reliability function, which is the probability *that the purchase time U is greater than a specific time ttt. Mathematically, R(t)=1−G(t), where G(t) is the cumulative distribution function (CDF) of the purchase time.*
- *t still represents a specific point in time.*
- *G(t) represents the probability density function (PDF) of the purchase time U.*
- *G(t) represents the distribution function of the purchase time U, i.e., the CDF.*
- *R(t)=0.5 indicates that 50% of customers have a purchase time greater than t. This is a median-related statement.*

The need for centralized analysis due to the normal distribution of customer purchase times remains the same, with G(t) and g(t) reflecting the distribution and density functions of the purchase time data.

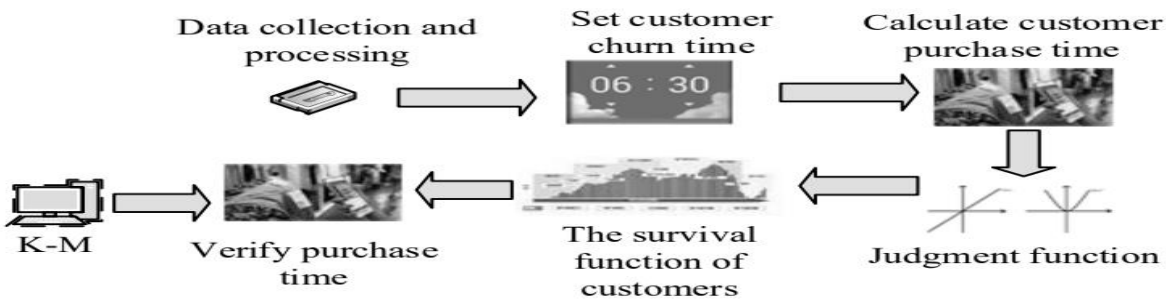

Figure 3: Probability Distribution of E-Commerce Business Data

In the analysis presented in Figure 3, the focus is on understanding and processing customer purchase time data. The steps involved in this analysis are as follows:

1. **Data Processing of Purchase Time**: The first step is to analyze and process the purchase time data of customers. This involves determining the current customer churn time, which is when customers stop purchasing, and then calculating the purchase time for each customer.
2. **Estimating Purchase Time**: By estimating the purchase time of customers, the customer churn function can be determined. This function helps in understanding the rate at which customers stop purchasing over time.
3. **Comparative Analysis**: The purchase functions of customers across different e-commerce platforms are compared. This comparison helps in drawing the customer survival function, which reflects the probability of a customer continuing to purchase over time.
4. **Validation of Current Purchase Time**: The survival function is then used to test and validate the current customer purchase time. This ensures that the data reflects the real-time purchase behavior of customers.
5. **Kaplan-Meier (K-M) Evaluation Method**: The evaluation and analysis of customer churn are typically done using the Kaplan-Meier (K-M) method. This is a classic non-parametric method, meaning it does not require assumptions about the distribution of customer purchase time. Instead, it focuses on the order of sample data to determine the probability of the current purchase time.
6. **Data Arrangement**: Assuming there are xxx customers in the sample, the data is arranged by the length of purchase time to create a new purchase time sequence. In this process, data that is neither deleted nor lost is retained, and the purchase time for each segment where data is lost or censored is calculated, as shown in equation (3) [19].

The Kaplan-Meier method is particularly useful in survival analysis as it deals with censored data (e.g., customers who haven't churned by the end of the observation period) and provides a step-by-step estimate of the survival function, which is vital in understanding customer behavior over time.

$$Q_r = \frac{m-r}{m-r+1} \dots\dots\dots Eqn\ 3$$

In equation (3), *r* represents the retained data that has not been deleted or lost. $Q_r$represents the probability of lost purchase time. *m* represents the total number of customer purchase times currently in existence. At this point, the size of the customer's purchase function value is equation (4)

$$C(t) = \prod_{t(r)<t} Q_r \dots\dots Eqn\ 4$$

In equation (4), $C(t)$ represents the customer's purchase time. When there are multiple non censored missing data, the variance of the data is calculated by estimating the minimum value of the purchase time function, and the estimation equation of the data is equation (5)

$$C(t) = \begin{cases} 1, t < t_{(r)} \\ \prod_{t(r)\le t} \frac{m-r}{m-r+1}, t \ge t(r) \end{cases} \dots\dots\dots Eqn5$$

In equation (5), when customer churn occurs, the function value corresponding to purchase time experiences a jump. The magnitude of this jump is determined by the number of customers lost at that particular time point, while ensuring that no two customers experience churn at the exact same moment. For customers with varying purchasing behaviors, these differences can be analyzed separately using statistical tests designed to identify production differences.

One commonly used method is the log-rank test, which is employed to compare survival distributions between groups. In this test, a distribution of the test statistics is constructed, and the sample statistic is compared to a critical value at a specified significance level. If the sample statistic exceeds the critical value, the initial hypothesis is rejected; otherwise, the initial hypothesis is retained.

The initial hypothesis assumes that the number of customers remaining is 1/m , where m is the total number of customers. The formula for the test statistic at this stage is given in equation (6) [22]

$$x^2(\log rank) = \frac{(L1-E1)^2}{E1} + \frac{(L2-E2)^2}{E2} \dots\dots Eqn\ 6$$

In the above equation

- **L1**: Represents the current number of e-commerce fans.
- **L2**: Represents the current number of non-fans.
- **E1 and E2**: Represent the current expected number of lost customers for the two categories (fans and non-fans), respectively.
- **X rank**: Represents the statistical quantity being tested, such as a test statistic in a statistical analysis.

This equation helps in determining the current customer churn situation through statistical testing. Additionally, dynamic analysis of customer consumption behavior on e-commerce platforms plays a crucial role in understanding customer patterns and preferences.

Figure 4 E-commerce Customer Goods Selection and Purchase Process

In Figure 4, the customer selection and purchasing process on e-commerce platforms is outlined through three primary methods. The first method is direct purchase, where customers select their products and place orders immediately without further actions. The second method involves customers collecting selected products and deciding whether to purchase them or leave the e-commerce site later on, reflecting a more deliberate shopping behavior. The third method consists of adding products to the shopping cart and returning to the e-commerce page at a later time to either complete the purchase or abandon the cart, indicating a more extended decision-making process. This variability in purchasing methods introduces uncertainty, particularly due to the potentially large time intervals between product selection and actual purchase, which affects the predictability of customer intent. Therefore, risk analysis of customer churn must take into account the dimensional information of customer characteristics. Figure 5 illustrates this dimensional information, which includes various aspects of customer behavior and attributes, providing a comprehensive understanding essential for analyzing and managing customer churn effectively.

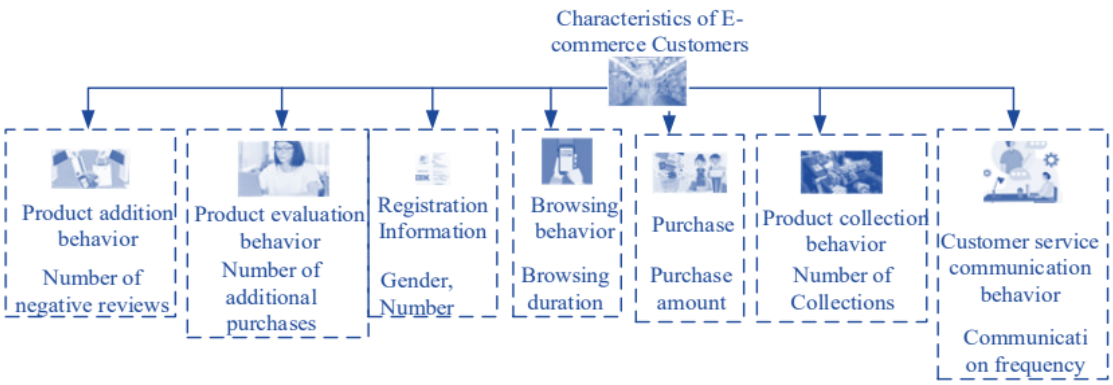

Figure 5 Customer Characteristic Dimension Information

In Figure 5, the changes in customer feature dimensions include customer registration information, browsing behavior, purchasing behavior, product collection behavior, customer service communication behavior, product evaluation behavior, and product addition behavior. These can all become risk factors for customer churn.

2.2 Building a CCR Evaluation Model Based on RF Algorithm and XGBoost

The risk of customer churn involves managing a substantial volume of data, and the problem often reduces to a secondary classification task. Analyzing customer churn risks requires thorough classification and analysis due to the inherent

complexity of the online shopping environment, which introduces greater variability in customer behavior compared to offline settings. As a result, the dataset size expands to accommodate diverse customer behaviors, necessitating prediction models that can handle large datasets efficiently while delivering high classification accuracy.

To address these needs, models such as Random Forest (RF) and XGBoost are selected for assessing risk factors. These models are well-suited for managing extensive data and providing robust performance in classification tasks. Specifically, Random Forest, which operates using Decision Trees (DT), employs various measures to gauge uncertainty in its predictions. Equation (7) provides the calculation formula for the uncertainty measure of Decision Trees within the Random Forest algorithm, which is crucial for understanding and interpreting the model's predictive reliability [24].

$$H(s) = \sum P(x) \log_2 P(x) \ldots \ldots . Eqn\ 7$$

In Equation (7), the terms are defined as follows:

- **S**: Represents all characteristic dimension data related to the current customer churn situation.
- **x**: Represents all currently occurring dimension types within the dataset.
- **p(x)**: Denotes the overall proportion of a particular dimension type xxx in the dataset.

Equation (7) is used to measure the uncertainty associated with the characteristic dimensions of customer churn.

In the context of risk assessment using Decision Trees (DT), the Gini index is employed as a benchmark for evaluating the effectiveness of splits within the tree. The Gini index helps to measure the impurity or purity of a dataset split and is calculated as follows:

$$Gini(T) = 1 - \sum_{i=1}^{n} pi2 \ldots \ldots . Eqn\ 8$$

Equation (8): Provides the calculation and judgment formula for the Gini index in Decision Trees. The Gini index is used to determine the quality of a split by calculating how mixed the target classes are in the resulting subsets after the split. Lower Gini index values indicate a purer split with more homogeneous subsets.

This approach allows for effective risk assessment and classification in customer churn analysis, utilizing Decision Trees and the Gini index to optimize model performance and accuracy.

In Equation (8), the terms are defined as follows:

- **Gini(T)** : Represents the Gini index, a measure of impurity used to evaluate the quality of splits in a Decision Tree (DT).
- **pi**: Denotes the probability of the current customer feature information appearing at type i at node t.

The Gini index is used to gauge the purity of the data at a particular node. A smaller Gini index value indicates higher purity of the data at that node, meaning that the data is more homogeneous with respect to the target variable.

In Random Forest (RF) models, the Gini index is crucial for determining how well a decision tree splits the data, with the average decrease in purity used for feature selection. The reduced impurity helps in selecting features that are more informative for the model. This approach significantly enhances the efficiency of the model by effectively handling large datasets and providing high classification accuracy. For improving computational efficiency and prediction accuracy, XGBoost is employed alongside RF. XGBoost excels in decision predictions due to its ability to handle large datasets, which is beneficial for e-commerce platforms with millions of customers. It also supports parallel processing to speed up the training process and is adept at managing sparse data, a common characteristic in e-commerce datasets.

Additionally, XGBoost offers various objective functions and evaluation metrics, allowing flexible adjustments and optimization according to specific business needs. It consolidates the results of the dataset through Decision Trees and utilizes them to make final predictions about customer churn, as represented in Equation (9) [27]. This combined approach of RF and XGBoost provides a robust solution for risk prediction and management in e-commerce environments.

$$\beta = \emptyset(x_i) = \sum_{k=1}^{k} f_k(x_i) \ldots \ldots Eqn\ 9$$

In Equation (9), the terms are defined as follows:

- β: Represents the predicted value size of Customer Churn Risk (CCR).
- $\emptyset(x_i)$: Denotes the total sum of customer risk datasets across K Decision Trees (DTs).
- $f_k(x_i)$: Represents the size of the data value within the number of K Decision Trees.

The equation integrates these components to predict customer churn risk by aggregating the results from multiple Decision Trees and considering the overall dataset characteristics.

The new dataset expression, given a set of sample data and feature size, is derived in Equation 9 [28]. This equation is used to reformulate the dataset based on the assumptions of the sample data and the feature set, enabling the updated risk prediction model to account for the refined dataset characteristics.

In XGBoost, to mitigate the impact of missing leaf nodes and splits caused by information gain in the model, it's essential to set specific values and directions for Decision Trees (DTs). This approach improves the computational efficiency of the algorithm. During feature training with DTs, if no features are missing, the algorithm operates normally. However, if features are missing during data prediction, the algorithm automatically directs the missing data to the appropriate branch of the Decision Tree. This mechanism helps maintain the model's accuracy and efficiency even in the presence of incomplete data [31].

The Customer Churn Risk (CCR) analysis for e-commerce involves considering multiple factors, as depicted in Figure 6. This comprehensive analysis includes various dimensions that impact customer behavior and churn, helping to create a more accurate and nuanced prediction model.

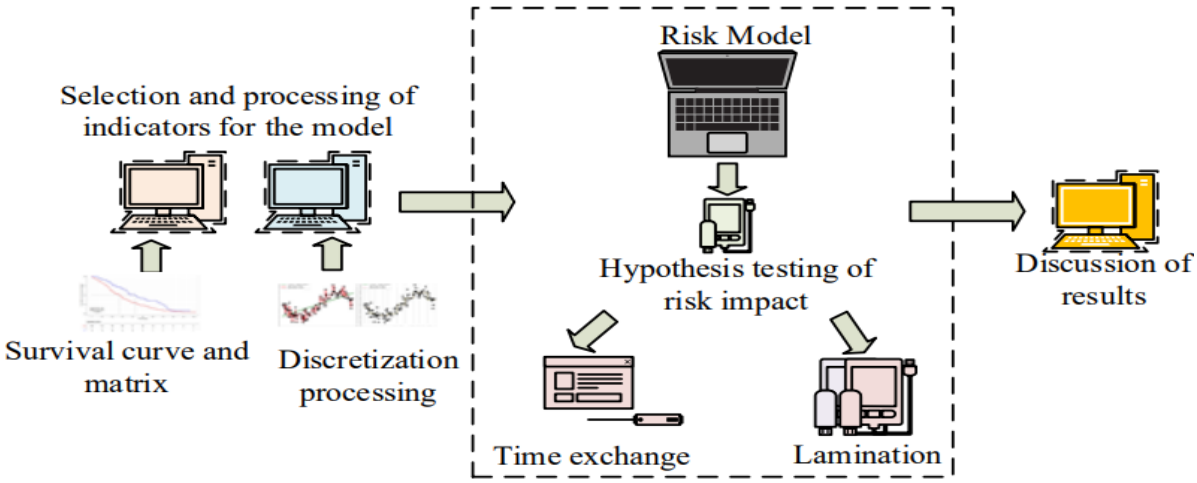

Figure 6 Analysis of Different Factors of Customer Loss Risk in E-commerce

In Figure 6, constructing a churn risk prediction model requires careful consideration of model indicators and their processing. This involves analyzing the survival impact curve of Customer Churn Risk (CCR), examining the scattered distribution of customer risk, and discretizing churn risk data. After setting up the model, the regression results are compared and discussed. The model includes hypothesis testing to evaluate the impact of various risk factors, regularization analysis to assess the effect of regularization on risk data, and hierarchical distribution analysis to understand the hierarchical nature of CCR. Finally, influencing factors are collected and predicted to maximize the effectiveness of customer retention strategies. For instance, by analyzing factors such as the probability of current customer churn, the likelihood of retaining customers, the value of customer churn, and the potential profit from successful retention, the model aims to predict and optimize the costs associated with customer retention [32]. The overall process of predicting customer churn risk in e-commerce, combining Random Forest (RF) and XGBoost, is depicted in Figure 7. This combined approach enhances predictive accuracy and improves the management of customer churn.

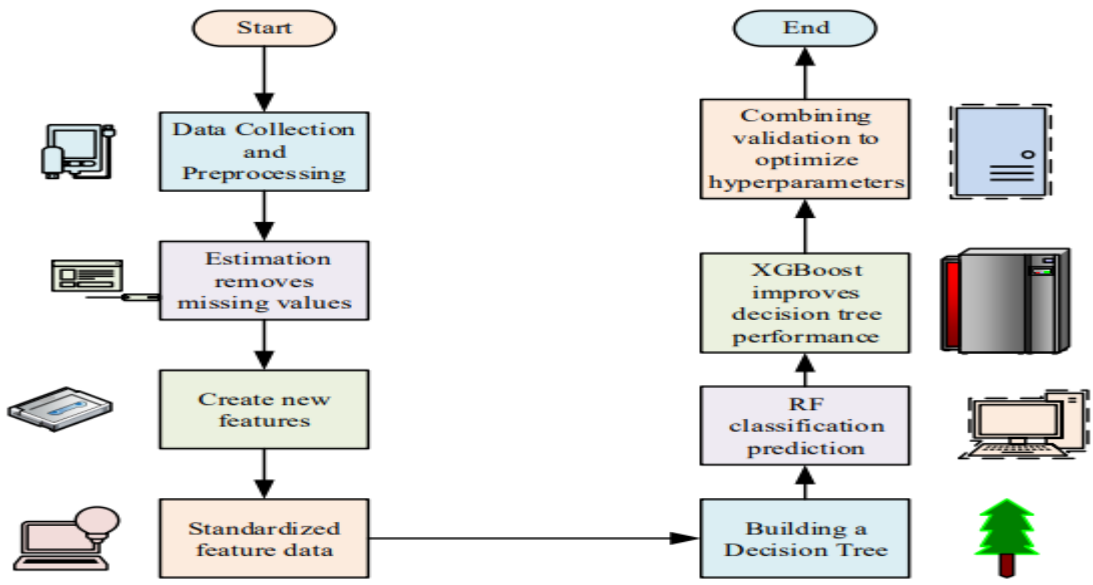

Figure 7 Data processing flow of e-commerce customer churn risk assessment model

3. CCR Model Performance Testing

This section focuses on analyzing Customer Churn Risk (CCR) for the current e-commerce platform by simulating the causes and probabilities of customer churn using various models. The performance of these models is then evaluated and compared against several traditional models to assess their effectiveness.

3.1 Analysis of CCR Prediction Results

In this research, an Intel Xeon E5-2690 v4 CPU with 28 cores and 56 threads, along with an NVIDIA Tesla P100 for image processing, is used. The system is equipped with 16 GB of HBM2 memory and 256 GB of DDR4 memory for the host. Ubuntu 20.04 LTS serves as the operating system, and Python 3.8 is chosen as the programming language. Apache Kafka is utilized for data collection and processing due to its efficiency in managing high-throughput, low-latency real-time data streams.

Apache Kafka functions as follows: it gathers data from diverse sources such as user behavior logs, transaction records, and customer service interactions from e-commerce platforms. Data is sent to specific topics within the Kafka cluster using the Kafka Producer API. Subsequently, the Kafka Consumer API retrieves this data from the Kafka topics and forwards it to downstream processing systems, such as Apache Flink, for real-time analysis.

The experiment focuses on maternal and child product data, using data from January 1, 2020, to December 31, 2021. It involves statistical analysis of purchases made by members and non-members in the same store over this two-year period. Data collected includes warehouse data, basic customer profile information, network behavior data, and browsing page data from a specific store. This data is used to predict customer churn risk (CCR), specifically analyzing the loss situation for babies in different age groups, as depicted in Figure 8.

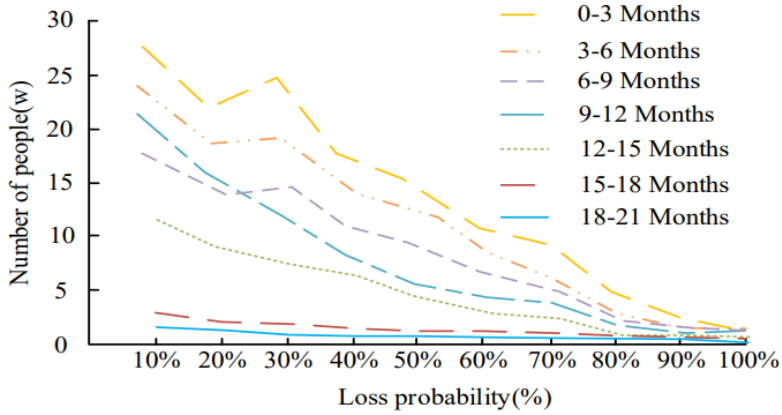

Figure 8 Probability of Loss of Mothers and Infants in Different Age Groups of Customers

In Figure 8, when the customer's baby is young, the overall risk of customer churn is relatively small, and the customer churn situation between different probabilities is basically small. There is a difference of approximately 23000 customers between 100% and 10% who experience customer churn in babies aged 0-3 months. This indicates that when the baby is young, customer churn is relatively small, and there is a greater dependence on e-commerce platforms, which is unlikely to cause customer churn. After the baby is older, such as within 6-15 months, the number of customers increases, and the number of people with different loss probabilities shows a downward trend. This may be due to an increase in the variety of products currently needed by the baby. After the baby is over 15 months old, there is a fluctuation in customer churn, which may be due to the current age group of baby customers having more selectivity in product selection. To compare the churn rates of customers with different purchasing power, the probability of customer churn with different purchasing power is compared to obtain Figure 9.

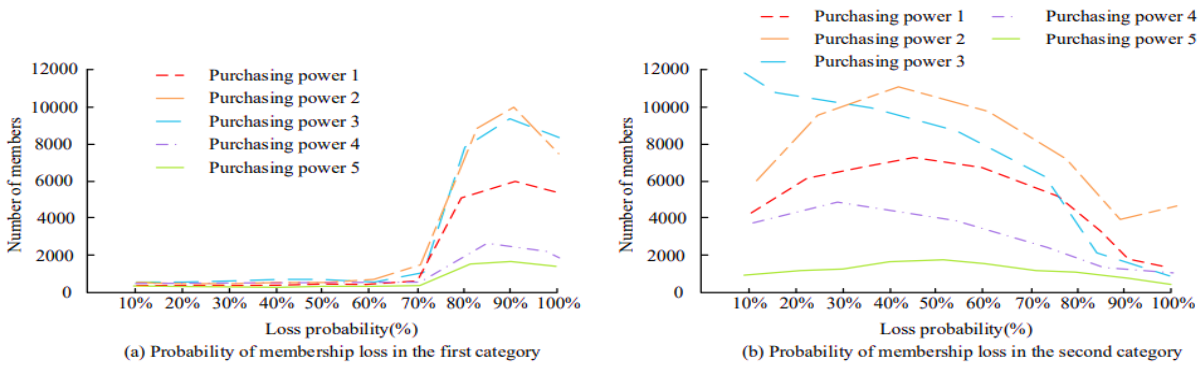

Figure 9 Comparison of Customer Churn Probabilities with Different Purchasing Power

The classification criteria for members are based on their purchasing behavior. Members are categorized according to their consumption level, with those meeting certain thresholds being classified into different membership types. Purchasing power is determined by the historical number of purchases made at the store, with higher purchasing power indicating stronger buying capability. In Figure 9(a), it is observed that there are relatively few customers with high purchasing power levels of 4 and 5, with the maximum number of such customers being only 1800. This suggests that

there are generally fewer customers with stronger purchasing power. Conversely, customers with lower purchasing power levels (1-3) exhibit a higher probability of churn, indicating that these members are more likely to leave, leading to an overall unhealthy churn situation.

Figure 9(b) shows that the probability of customer churn for members with purchasing power levels of 1-2 follows a parabolic trend, peaking at a churn probability of 50%. This indicates that customers within this purchasing power range exhibit higher mobility and a more noticeable engagement with e-commerce platforms. In contrast, the number of members with high purchasing power levels of 4-5 is relatively small, and these customers have a lower likelihood of churn. This suggests that customers with higher purchasing power have a stronger dependence on the store and are less likely to experience turnover. To assess the impact of negative reviews on customer churn, Figure 10 presents a comparison of negative reviews across different member categories. This analysis helps determine how negative feedback influences customer retention and churn rates.

3.2 Model Result Analysis

To test the performance of the current research model, the running time and Root Mean Square Error (RMSE) of the model are compared. Figure 11 tests the XGBoost algorithm and RF algorithm. RMSE-NAN/RMSE-unNAN and Time-NN/Time-unNAN represent the RMSE and training runtime of the model missing data/after filling in missing value data

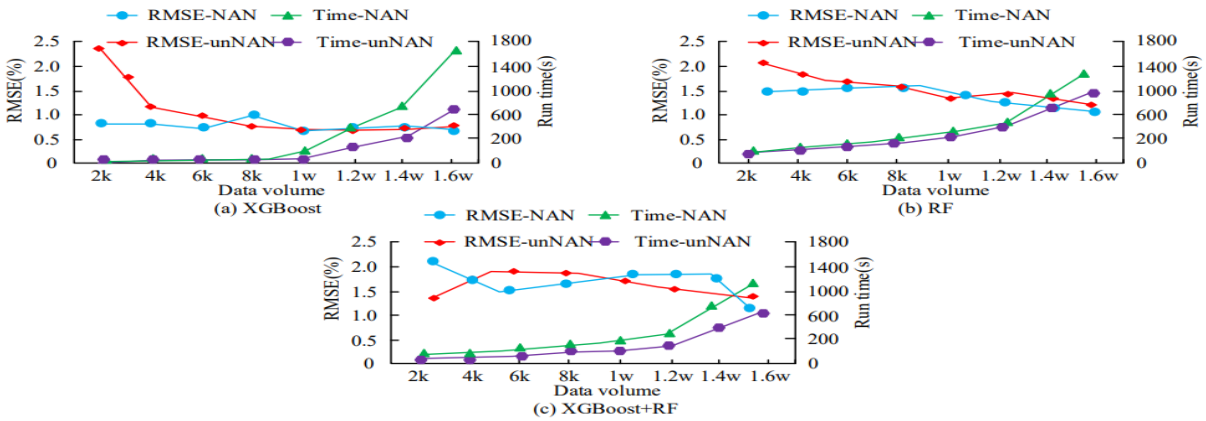

Figure 10. Algorithm Model Test Results

In Figure 10(a), as the amount of missing data increases, the RMSE value tends to stabilize, indicating that imputing missing data enhances the model's accuracy. The consistency between the runtime of the training process and the time required for data imputation further supports this. In Figure 10(b), post-imputation, the resulting curve becomes smoother, and the variations in the training data are minimized. In Figure 10(c), when two models are employed for training, the combined approach demonstrates a reduction in both time cost and error rate compared to training a single model, suggesting improved accuracy and efficiency. The dual-model approach outperforms the single-model method. To evaluate the performance of the models in use, comparisons were made with other data models such as Logistic Regression (LR) and Light Gradient Boosting Machine (LGBM), with the prediction target being a sample size of 2,000 individuals. The research model achieved a prediction accuracy of 94.6%, surpassing the lowest accuracy of 88.4% obtained by the LR model by approximately 6.2%. Additionally, the recall rate of the research model is 7.5% higher than that of the LR model. These findings indicate that the research model is more effective and better suited for predicting customer churn. The lower predicted churn count also suggests that the research model delivers higher accuracy when predicting large datasets. Figure 11 illustrates a comparison of the training runtimes across the aforementioned models.

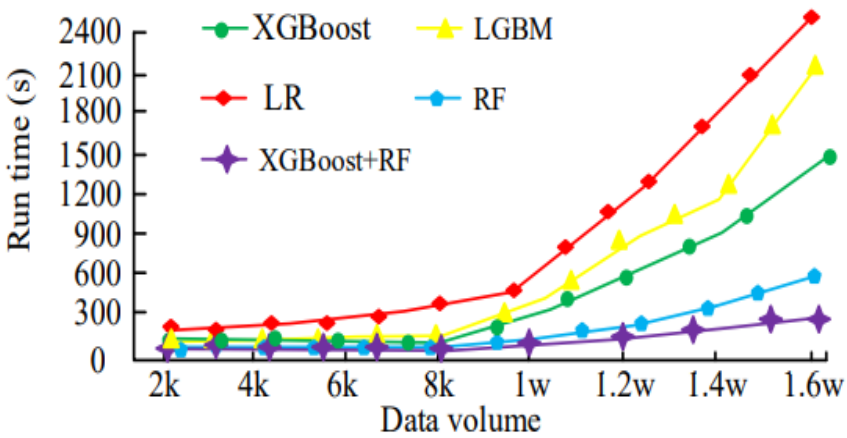

Figure 11 Comparison of Running Training Time for Different Models

In Figure 11, the training time for the research model remains consistent, stabilizing around 300 seconds as the data volume increases, demonstrating its efficiency and stability. When compared to other models, the research model exhibits a significantly shorter training time. Specifically, the average training time for the research model is 346 seconds, which is notably lower by 196 seconds compared to the 542 seconds required by the Random Forest (RF) model. It is also 671 seconds shorter than XGBoost's 1017 seconds, 1203 seconds less than the 1549 seconds required by Logistic Regression (LR), and 940 seconds shorter than the 1286 seconds for the Light Gradient Boosting Machine (LGBM). Overall, the research model not only offers a reduced training time but also delivers superior performance.

4. Discussion

In today's competitive e-commerce landscape, customer churn presents a significant challenge for organizations, leading to not only direct revenue loss but also potential harm to brand reputation and market share. Traditional methods of assessing customer churn risk typically rely on simple statistical analysis and rule-based models, which often struggle to handle large-scale, high-dimensional, and complex user behavior data. This limitation hampers their ability to deliver accurate and timely predictions.

With advancements in machine learning and data mining technologies, there has been a growing trend toward applying advanced algorithms to enhance the accuracy and robustness of customer churn predictions. This research introduces a novel model that combines the XGBoost and Random Forest (RF) algorithms for assessing customer churn risk. The study reveals that churn risk varies across different age groups, with older customers exhibiting a higher risk of churn, potentially due to differences in age and store merchandise preferences. When analyzing churn among customers with varying purchasing power, the research shows that customers with higher purchasing power have lower churn rates, likely due to greater trust in the store. Conversely, customers with lower purchasing power exhibit higher churn rates. Additionally, the analysis of negative store reviews indicates a direct correlation between the increase in negative reviews and the rise in customer churn, suggesting that negative feedback significantly impacts store operations and customer purchasing decisions.

The research also compares the performance of different algorithmic models under varying levels of missing data. The findings suggest that as missing data increases, the RMSE value stabilizes, and the model's accuracy and runtime become more consistent, likely due to the influence of missing values on the model's predictions. The study's model outperforms others in terms of accuracy, with a top accuracy of 94.6%, representing a 6.2% improvement over the lowest-performing Logistic Regression (LR) model. The recall rate of the study's model is also 7.5% higher. These improvements are attributed to the use of a combined model that leverages the strengths of both RF and XGBoost. In terms of training time, the research model demonstrates efficiency, with an average training time 196 seconds shorter than the RF model, 671 seconds shorter than XGBoost, and 1203 seconds shorter than LR. This efficiency may be due to the model's ability to effectively manage hyperparameters through XGBoost. The study also highlights the strengths of RF in handling large-scale datasets by constructing multiple decision trees for parallel computation, maintaining stable performance even as the number of features increases. This capability allows RF to efficiently process large amounts of user behavior data from e-commerce platforms during the feature screening stage, providing concise and effective feature sets for subsequent modeling. Meanwhile, XGBoost excels in handling large-scale data and high-dimensional features by gradually optimizing model parameters, quickly adapting to different datasets and business requirements. Its built-in regularization mechanism helps prevent overfitting, allowing the model to maintain high prediction accuracy and robustness despite changes in data size and complexity.However, the study acknowledges some limitations. Data from e-commerce platforms often contain noise, incompleteness, and outliers, which can affect the model's accuracy and robustness. Furthermore, the process of feature data processing is complex, time-consuming, and reliant on expert analysis, making it difficult to automate and potentially hindering the study's efficiency.

Despite these challenges, the study's model demonstrates strong generalization and adaptability across multiple e-commerce platform datasets, indicating its potential for broader application. Through several experimental validations, the model proves to be highly accurate and robust in predicting customer churn, underscoring its effectiveness under controlled variables and disruptive factors. In conclusion, the research model effectively combines the strengths of RF and XGBoost, capturing the complex features of user behavior and proving its applicability in customer churn risk assessment. The model's design ensures structural validity and provides a strong foundation for robust and stable customer churn predictions, particularly in handling large feature-rich datasets. Compared to other approaches, such as SVM, neural networks, and deep learning, RF and XGBoost offer a balanced approach with superior results in e-commerce customer churn analysis.

5. Conclusion

This study focused on analyzing customer churn in e-commerce using Random Forest (RF) and XGBoost models to provide a more accurate understanding of churn behavior in stores. The experimental findings revealed that customers with higher purchasing power exhibited greater variability in churn, while those with lower purchasing power experienced more stable churn rates. Additionally, negative product evaluations significantly influenced customer churn, with a higher likelihood of customer loss associated with an increase in negative reviews. When comparing model performance, the research method achieved an RMSE of 2.1, approximately 1.4% lower than the highest RMSE of 3.5 observed with the Light Gradient Boosting Machine (LGBM). The research model also demonstrated the highest accuracy at 94.6%, which is 6.2% higher than the lowest accuracy of 88.4% obtained by the Logistic Regression (LR) model. The recall rate of the research model was similarly superior, showing a 7.5% improvement over the LR model. In terms of training time, the

research model remained stable around 300 seconds, with an average training time of 346 seconds. This was significantly lower than the training times of other models: 196 seconds shorter than RF, 671 seconds shorter than XGBoost, 1203 seconds shorter than LR, and 940 seconds shorter than LGBM. Furthermore, the RMSE value for missing data in the research model remained stable, reinforcing the model's robustness.

In summary, proposed models demonstrated superior predictive performance in forecasting customer churn, with most churn instances being attributed to limitations in customer service, inadequate after-sales support, and negative store reviews. Future research will expand predictive analysis across different stores to uncover additional factors contributing to customer churn, aiming to enhance the operational efficiency of current stores. While this study combined only two models, it did not analyze the temporal patterns of customer behavior. Therefore, future research will incorporate models such as recurrent neural networks (RNN) and long short-term memory (LSTM) networks for more comprehensive analysis. Additionally, further studies will explore larger datasets to refine and expand the research findings.

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